

SHARIF UNIVERSITY OF TECHNOLOGY

Department of Electrical Engineering

PROJECT ASSIGNMENT

Large-Scale Traffic Pattern Mining and Detection-Consistency Analysis

INSTRUCTOR / INSTRUCTORS

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ACADEMIC CONTEXT

Big Data Processing

Contents

1 Scenario and independent deliverables 2

2 Data dictionary and identifier interpretation 2

3 Part 1: frequent patterns, registration-quality heuristics, and LSH 2

4 Part 2: five entity-clustering scenarios 3

5 Part 3: streaming architecture and API presentation 3

6 Part 4: movement graphs, PageRank, HITS, and comparison 4

7 Complete operator-code reference 4

8 Complete plate-letter code reference 5

1 Scenario and independent deliverables

Act as a data-analysis engineer for a road authority. Use vehicle-passage records to investigate registration quality, traffic patterns, clustering, streaming, graphs, and anomalies. Minimum implementation does not guarantee the top relative grade: justify the problem, assumptions, parameters, quality checks, visualization, and decision relevance. For each of the four parts provide runnable documented code, configurable parameters, meaningful visualization, a separate engineering report, and a presentation video no longer than ten minutes. Thus the course requires four independent reports and four videos. Each report explains problem/assumptions, data/preprocessing, model/parameter design, results/insights, validation and sensitivity, limitations, and extensions. README documents execution, inputs/outputs, dependencies, and key parameters. A video explains purpose, methods, functions or endpoints, interpretation, and limitations. The supplied files determine which deliverables were actually archived; do not infer missing videos from these requirements.

2 Data dictionary and identifier interpretation

CompanyCode denotes operator/contractor; DeviceID identifies a camera; Lane identifies its lane; PassedTime is the passage timestamp; ReceivedTime is system receipt; event_ts uses parsed PassedTime and otherwise the declared receipt-time fallback. PlateType and VehicleType are category codes; Speed is recorded speed; road_code/road_name identify the road; state_name/state_code identify the province. StreetID, PassedVehicleRecordsId, and SystemCode are internal/system identifiers. The assignment describes nine-digit encoded plate values with a two-digit letter code. Other digits have been consistently indexed for privacy. Treat them as pseudonymous identifiers, not recoverable raw registration numbers. Code 21 represents the letter Ain, suggested as a heavy-fleet proxy. A proxy is not guaranteed ground-truth vehicle classification. Keep identifiers as strings when leading zeros matter.

3 Part 1: frequent patterns, registration-quality heuristics, and LSH

Load and preprocess the data and optimize access with a B-tree-based index as requested. Consider (PlateValue,event_ts), DeviceID, or a justified alternative. Measure at least one relevant operation before/after indexing, such as basket construction, route preparation, or exact Jaccard computation. Explain when the index helps and when it does not. A columnar file layout can be an alternative experiment but is not literally the requested B-tree. Define BasketID=(PlateValue,Window(event_ts,delta_t)). Try a short interval such as 30 minutes and one hour; justify the choice and examine sensitivity. Define items as (CompanyCode,DeviceID), or compare a stated alternative. Deduplicate items within a set-based basket. Use Apriori or PCY as specified and extract/interpret frequent triples with a justified minimum support. A triple is co-occurrence, not an automatically established ordered route. Use frequent patterns to identify potentially missing registrations. State how an expected observation and denominator are formed, compute camera-level miss rates, aggregate to company level, and compare traffic volume with rate. Consider speed bands, lane, and time effects. Explain possible alternate routes and incomplete observation before declaring hardware failure. For each vehicle extract a set of road codes, generate MinHash signatures, use LSH banding to propose similar pairs, and compute exact Jaccard for candidates. Report signature/band parameters, sample restrictions, and thresholds. Compare miss heuristics inside similar-route groups with the overall rate. Sets do not preserve traversal order, travel time, or lane; matching them does not prove an identical physical trip. Deeper analysis and parameter sensitivity form part of evaluation.

4 Part 2: five entity-clustering scenarios

1. Camera behavior: aggregate by DeviceID, including 24-hour temporal profile, speed mean/dispersion, heavy ratio, and weekly stability. Normalize to avoid scale domination, use PCA, compare K-Means initializations, and interpret roles such as transit/night, morning commute, or uniform traffic.

2. Lane behavior: use (DeviceID,Lane), with temporal profile, speed, fleet composition, and lane share of camera traffic. Normalize, reduce dimension, cluster, and explain within-camera differences.

3. Road behavior: aggregate all cameras on road_code or road_name. Include time distribution, speed, fleet composition, and plate diversity. Interpret road groups and discuss policy relevance without claiming that clustering alone selects an optimal intervention.

4. Vehicle behavior: use hashed/pseudonymous MasterPlateValue, require sufficient records, and describe usual hours, road diversity, speed, and vehicle-type proxy. The limited one-week horizon and large identity count require careful sampling and noise thresholds.

5. Camera data quality: aggregate receipt-minus-passage delay, invalid PlateType, missing fields, and temporal instability. Group devices by observed quality profiles and propose investigation priorities. Validate any diagnosis against external maintenance evidence.

5 Part 3: streaming architecture and API presentation

Select the busiest day by PassedTime/event date. Extract its records, order by ReceivedTime, and replay them line by line through a socket producer. Make rate configurable and compare several rates, with the assignment suggesting examples around 50, 100, and 1000 records per second. Messages include DeviceID, PlateValue, event_ts, ReceivedTime, Speed, road_code, and state_name. Preserve original timestamps independently of replay wall time. Choose a window duration, such as one or five minutes, and a slide. Provide structured API endpoints for every implemented analysis and a separate visualization program that calls them. FastAPI is suggested but not mandatory. Document endpoint fields, units, freshness, and errors. Compute total and per-camera counts, and the three busiest cameras per window. Explain the need for aggregation, the sensitivity/stability tradeoff, and short/long-window effects. Analyze receipt-minus-event delay with mean, maximum, and distribution; compare event-time and receipt-time windows and late-data handling. Compare at least two window lengths. Examine camera shares, unusually dominant cameras, and rank stability. Compute average speed and study its relationship to counts. Define any congestion indicator explicitly and discuss its limits. Identify busiest roads and provincial patterns/hotspots. Keep geographic aggregation distinct from a validated road-density measurement. Maintain previous-window state and cumulative counts since stream start. Ensure overlapping-window records are counted only once in true cumulative totals; repeated window updates need stable identities or equivalent idempotent logic. Maintain an online EWMA mean/variation baseline per camera. Calculate a drop anomaly score and require L consecutive abnormal windows before alarming. Return camera, window, observed count, baseline mean/variation, score, and alarm status. Include zero-count windows for monitored inventory so fully silent cameras are visible. Design a health index from stability, fluctuations, and anomalies with justified thresholds for healthy/suspicious/critical. Forecast the next window from past windows, compare prediction and actual count, and report quantitative error across a sufficient sequence. A forecast or alert is not a validated diagnosis merely because the dashboard displays it.

6 Part 4: movement graphs, PageRank, HITS, and comparison

Sort each vehicle's observations and form consecutive transitions. Build directed camera, road, and province graphs, with optional additional models. Report node/edge counts, in/out degree or weighted strength, and prominent links. Define structural anomalies precisely, such as changes in common links, unusual travel times, or incomplete next observations. Select meaningful subgraphs and visualize them. Run PageRank on all three graphs, explain what weighted structural importance means at each level, incorporate rank into explicitly defined anomaly criteria, and visualize node/link importance. Specify damping, teleportation, weights, dangling nodes, convergence, and any custom link-importance formula. Run HITS on the camera graph and interpret hub/authority roles. Define a bipartite model such as vehicle-to-camera, analyze its HITS output, and compare the information it provides with consecutive-camera movement. Define role-based anomalies and visualize ranks/roles. Compare structural, rank-based, and role-based anomaly results and their limitations. Missing observations, routes, sampling, one-way infrastructure, and timing gaps can all produce apparent asymmetry.

7 Complete operator-code reference

Operator names are rendered in English transliteration. Contract descriptors distinguish existing/new and provincial/headquarters entries; identifiers are unchanged.

CompanyCode	Operator	Contract descriptor
118	Rahkam Communications	Upgraded existing; provincial
116	Hooshmand Dayegan	Existing; provincial
183	Roshan Tolou Shargh	Existing; provincial
102	Arg	Existing; provincial
100	Pooyesh Pardaz	Existing; provincial
105	Negaresh Rayaneh	Existing; provincial
205	Fard Iran	Provincial
117	Rahkam Communications	Upgraded existing; provincial
123	Basir Pardazesh	New; headquarters
121	Tozin Electric	Provincial
206	Fanap	New; provincial
313	Paya Ravesh Malagar	New; provincial
181	Fard Iran	Headquarters
122	Negaresh Rayaneh	New; headquarters
133	Arg	Not specified
131	Ariana	New; headquarters
129	Sairan	New; headquarters
184	Badi Isfahan Intelligent Systems (Bistco)	Existing; provincial
207	Farasoo Tozin	Provincial
136	Hamsoo	New; headquarters
112	Fanap	Existing; provincial
141	Fanap Infrastructure	New; headquarters
140	Rahkam Iranian Communications	New; headquarters
115	Pand Industries	Not specified
110	Taba Tozin	Not specified
168	Isiran	New; headquarters
169	Aria Hamrah	New; headquarters

CompanyCode	Operator	Contract descriptor
173	Basir Pardazesh	New; provincial
172	Middle East Alpha Processing	Not specified
177	Gostaran Sanaye Novin Pars	Provincial
178	Sairan	New; provincial
176	Arg	New; provincial
182	Asr Rayaneh Parseh Iranian	New; provincial
174	Roshan Tolou Shargh	New; provincial
208	Badi Isfahan Intelligent Systems (Bistco)	New; provincial
314	Sepehr Sharif Intelligent Systems	Not specified
220	Delta Pars	New; provincial

8 Complete plate-letter code reference

The 34 codes below retain their two-digit identifiers. English letter names describe the encoded symbols; D and S are literal Latin-letter categories. The remaining plate digits are pseudonymous indexes and do not recover an actual registration number.

Letter code	English symbol name
01	Alef
02	Beh
03	Peh
04	Teh
05	Theh
06	Jim
07	Cheh
08	Heh (Arabic)
09	Kheh
10	Dal
11	Zal
12	Reh
13	Zeh
14	Zheh
15	Sin
16	Shin
17	Sad
18	Dad
19	Ta
20	Za
21	Ain
22	Ghain
23	Feh
24	Qaf
25	Kaf
26	Gaf
27	Lam
28	Mim

Letter code	English symbol name
29	Nun
30	Vav
31	Heh
32	Yeh
33	D
34	S